Disclosure Regulation, Stock Liquidity, and International Policy Diffusion

Abstract

We examine stock liquidity around public-disclosure requirements and the international spread of these requirements. The calendar contains 269 country–family topic milestones in 66 countries during 2000–2025, including EU climate reporting. After removing overlapping NFRD topic tags from the pool, the three-year comparison supports 6 families, 64 implementations and 47 countries. Equal-family stacked estimates correspond to 10.2% lower trading costs ($p = 0.268$); the market-cap-weighted family average corresponds to 17.1% lower trading costs ($p = 0.043$). The EU-inclusive climate comparison covers 14 adopting countries. We then examine learning, copying, competition and external finance. Adoption is positively associated with favourable growth among earlier adopters in the equal-family hazard model, while liquidity-based learning is imprecisely estimated. The evidence indicates differences across reporting families and pooling targets rather than a uniform market response or an established diffusion mechanism.

Keywords: disclosure regulation; stock liquidity; policy diffusion; international reporting.

1 Introduction

Disclosure rules shape what investors learn about companies. Public information can narrow differences between informed and uninformed investors and reduce the cost of trading (Diamond and Verrecchia 1991, Leuz and Verrecchia 2000). Governments now require companies to report on matters ranging from financial statements and audit risks to emissions, taxes and supply chains. Research on IFRS and ESG reporting links these requirements to capital-market outcomes, while showing that enforcement and reporting incentives matter (Daske et al. 2008, Christensen et al. 2013, Krueger et al. 2024).

Much of this evidence concerns a particular reporting requirement. We ask whether disclosure rules, considered together, are associated with greater liquidity in adopting countries. We also ask

why the rules spread. Countries may learn from outcomes abroad, follow familiar examples, respond to competitors or face external pressure. Koning et al. (2018) distinguish these explanations for international IFRS adoption. We apply that framework across disclosure families and separate exposure to a foreign rule from the outcomes observed after its implementation. Figure 1 sets out the question, reporting families and related studies.

We combine Datastream company records for 90 countries during 2000–2025 with a calendar of 269 country–family reporting milestones in 66 countries. The calendar brings together financial reporting, auditor reports, ESG and climate information, public country-by-country reports, extractive payments, supply-chain statements, sustainability standards and electronic reporting. Laws, official notices and published country lists establish the content and timing of each requirement. Our primary liquidity comparison follows the same companies before and after 64 implementations in 47 countries, across six families. Countries with documented later implementation provide the comparisons.

Trading costs are lower after implementation under both family weighting schemes, but the estimated magnitude and precision differ. Equal-family stacking gives a 10.2% decline ($p = 0.268$); the market-cap-weighted average of family estimates gives a 17.1% decline ($p = 0.043$). The separate climate comparison, including EU reporting, covers 14 adopting countries and indicates 18.0% lower trading costs ($p = 0.086$). These are averages across different reporting requirements and comparison samples. They do not establish a common liquidity improvement for every family.

The diffusion tests distinguish learning, copying, competition and external finance. Favourable growth among earlier adopters is positively associated with implementation in the equal-family hazard model (coefficient 0.936, $p = 0.026$). The interaction between foreign-policy exposure and earlier adopters' liquidity improvements is positive but imprecise (0.277, $p = 0.406$). Neither result survives the full multiple-testing adjustment. Together, the two parts connect the market consequences of disclosure rules to their international spread while retaining the differences in reporting content, implementation timing and empirical support.

2 Data and Research Design

2.1 Sample and datasets

Our market data come from Datastream and cover 90 countries during 2000–2025. We construct the policy calendar from statutes, standards, official notices and published country lists. The source archive contains reviewed legal provisions for 81 market-data countries, together with the original documents, amendments and successor rules. The Online Appendix provides the country lists and source detail.

The public-disclosure calendar contains 269 dated country–family topic milestones in 66 countries during 2000–2025 (Table 1 and Figure 2). Twenty NFRD climate tags overlap an ESG milestone and are excluded from pooled estimation under climate. The inventory records reporting phases rather than an exhaustive census of first-ever laws. The full risk panel covers 82 countries including confidential tax reports and 67 for public reporting. The source extension matches all 38 Krueger treatment countries and records all 32 Rauter jurisdictions; 31 of the latter belong to our market universe. Published adoption years, legal force and first covered fiscal periods remain distinct (Krueger et al. 2024, Rauter 2020).

For each requirement, we record its legal scope, audience and implementation date. Legal adoption, entry into force, the first covered financial period and the filing deadline are separate dates. A mandatory report can allow materiality judgements or explanations for individual items. Public disclosures form the main sample; confidential tax returns supplied to tax authorities are analysed separately. The calendar retains the date supported by the source and distinguishes a statutory deadline from an observed filing.

The market data provide three liquidity proxies following the reporting literature (Amihud 2002, Lesmond et al. 1999, Krueger et al. 2024). *Bid-Ask Spread* measures the cost of trading immediately; lower values indicate greater liquidity. *Price Impact* measures the association between absolute returns and dollar trading volume. *Zero Returns* is the fraction of observed trading days without a price change. We use log trading costs and log price impact, screen for sufficient annual

observations, and winsorize the first two measures at the annual first and ninety-ninth percentiles. An additional illiquidity index averages the three standardized measures. Higher values of each outcome indicate less liquidity.

World Bank data provide income, GDP growth, trade, aid and IMF credit measures. IMF bilateral portfolio positions measure financial exposure to foreign markets. Fixed geographic, language and export-pattern networks identify other peer groups. These public country data complement the licensed market observations; the firm-level Datastream files are not publicly redistributed.

2.2 Liquidity comparisons and weights

We form a separate comparison for each supported family and implementation cohort. For a milestone in year g, the primary window contains years $g-3$ through $g-1$ and $g+1$ through $g+3$. We omit year g as a transition year. Comparison countries have a documented later milestone after the outcome window. An unknown date does not supply an untreated observation. This construction avoids using already implemented cases as later-adopter controls for the same series, while recognising that countries may have other or earlier reporting requirements (Goodman-Bacon 2021, Callaway and Sant'Anna 2021, Sun and Abraham 2021, Baker et al. 2022).

We choose each company's largest positive-market-value security line in year $g-1$ and hold the companies fixed over the window. A country must supply at least 30 qualifying companies with the required annual market measures. This gives a comparison among firms with persistent observable trading data and limits changes in company composition. It also reduces country coverage and can favour larger, more actively traded companies. Two-year windows assess this restriction with a different horizon; they do not provide an identical sample.

Within each cohort we estimate a country- and year-effect comparison, with country weights based on GDP in $g-1$. Cohort weights within a family reflect the baseline GDP represented by adopting countries. The primary sample contains 1,704 country–stack–years, 47 countries, 64 country–family implementations and 21 family–cohorts. The full climate comparison is reported separately because the same NFRD reform also contains ESG disclosure. Country-cluster scores

retain dependence when a country appears in several cohorts or families.

The main presentation gives each supported family equal objective mass. We report a stacked common-coefficient regression and an explicit average of separately estimated family coefficients. The Online Appendix also reports market-cap family weights based on the baseline sample market value represented by adopting-country companies. These are different economic targets, not interchangeable estimates. The sources and pooling rule are fixed before inspecting the revised results; the complete coefficient registry retains both weights and every declared specification.

The secondary specifications add lagged log income per capita, GDP growth and trade/GDP; use two-year windows; and examine alternative liquidity proxies. Event-time coefficients show differences before and after implementation. These checks assess the comparison, but cannot establish parallel trends or remove all coincident reforms (Roth et al. 2023, Rambachan and Roth 2023).

2.3 International implementation tests

We estimate implementation-hazard models with family and year effects, and linear probability models with country–family and family–year effects. The latter absorb persistent country-family differences and family-specific international waves. Independent national choices form the main comparison; coordinated implementations enter a separate comparison. All signals use the preceding year or earlier. These models explain documented implementation timing. Because the political decision can precede implementation, they do not establish that governments used a signal before making their original decision.

Portfolio exposure is the share of observed foreign portfolio positions in countries that have implemented the requirement. We require documented histories to cover at least half of the observed portfolio mass. The liquidity-success signal measures lower trading costs among the same companies in earlier adopting countries. A donor must have implemented by $t-2$; its observed change ends in $t-1$. This is an observed foreign outcome, not an estimated liquidity improvement caused by the foreign policy. We include exposure, success and their interaction together.

Following Koning et al. (2018), growth learning compares prior-year growth among earlier and later adopters in a geographic network with documented histories. Copying uses a larger geographic peer fixed with GDP in 2000, distinguishing recent and older implementation; language peers provide a second proxy. Competition uses export-pattern peers selected in 2000. Financial dependence adds aid to IMF credit and SDR allocations relative to GDP. It does not identify a lender's policy conditions. Missing histories, portfolio positions and outcomes remain missing.

Both family weights are applied to the same sample for each comparison. We require at least 15 countries and sufficient implementation and nonimplementation observations. Unstable or unsupported models remain in the full results registry. The finite core matrix contains 160 fits across channels, audiences, implementation types, estimators and weights. We report country-cluster inference and retain the Holm adjustment across all 256 attempted focal tests (Holm 1979). The channel estimates are exploratory.

3 Liquidity Findings

3.1 Liquidity around implementation

Table 2 and Figure 3 report the equal-family comparisons. The stacked coefficient is -0.108 ($p = 0.268$), corresponding to 10.2% lower trading costs. The explicit family average is -0.061 ($p = 0.378$), corresponding to 5.9% lower trading costs. Both confidence intervals include zero. Under market-cap family weights, the stacked coefficient is -0.176 ($p = 0.071$) and the explicit family average is -0.187 ($p = 0.043$), or 17.1% lower trading costs (Online Appendix Table OA6). Weighting therefore affects the estimated average and its precision.

The six-family comparison includes expanded auditor reports, public bank reports, climate reporting, ESG reporting, IFRS and supply-chain disclosure. Family estimates differ (Online Appendix Table OA1). Adding documented later IFRS implementation supplies an additional comparison country and supports a New Zealand cohort that did not meet the earlier comparison-country requirement. That single-adopter IFRS contrast is positive, while several other family

estimates are negative. The pool should therefore be read as a weighted average across supported comparisons, not evidence that each type of disclosure improves liquidity.

3.2 Climate reporting and EU coverage

The full climate comparison includes 14 adopting countries and 20 countries overall. Its trading-cost coefficient is -0.199 ($p = 0.086$), corresponding to 18.0% lower trading costs, with a 95% interval from -0.429 to 0.031 in logs (Online Appendix Table OA18). Adopting countries are Austria, Belgium, Denmark, Finland, France, Germany, Greece, Italy, the Netherlands, Poland, Spain, Sweden, the UK and Vietnam. NFRD reporting is materiality-based environmental disclosure; France's earlier inventories and the UK and Vietnamese rules have different content and perimeters. The estimate concerns this broad climate-reporting topic and cannot isolate a single emissions metric.

NFRD observations already represented by an ESG milestone are omitted from the pooled climate component to avoid counting the same reform twice. They remain in the separate climate estimate and in the complete source calendar. The pooled climate row consequently has three adopting countries: France, the UK and Vietnam. The broad calendar contains 34 climate jurisdictions; recent milestones lack the post years or documented comparison countries required by the design. The supply-chain comparison retains Australia, France, the UK and the US market observation based on California's rule, which is a subnational eligibility-based requirement rather than a federal obligation for every issuer.

3.3 Interpretation and supporting comparisons

Country-condition specifications, shorter windows, alternative liquidity measures and event-time estimates appear in the Online Appendix. Payments-to-governments coverage is substantially wider after reconciliation with Rauter (2020), but this family still lacks three qualifying later-adopter comparison countries for an eligible cohort under the fixed design. More legal coverage therefore does not automatically produce an identified liquidity comparison. Confidential tax re-

porting remains outside the public-disclosure average.

Pre-implementation patterns, persistent-company selection, concurrent reforms and partial legal coverage limit causal interpretation. Country clustering preserves statistical dependence across stacks, but cannot separate simultaneous legal requirements or make a materiality-dependent report equivalent to a universal metric mandate. We interpret the results as country-market comparisons around documented reporting phases.

4 International Policy Diffusion

4.1 Diffusion Mechanism 1 – Learning from observed outcomes

Table 3 relates implementation to foreign-policy exposure and observed liquidity among earlier adopters. The equal-family exposure–success interaction is 0.277 ($p = 0.406$) in the hazard model, using 270 country–family–years in 26 countries. The market-cap counterpart is 0.422 ($p = 0.179$). The within-country equal-family interaction is 0.010 ($p = 0.647$). These estimates do not establish that governments respond to liquidity gains abroad; the full focal-test adjusted p-values are one.

Following Koning et al. (2018), we also compare economic growth among earlier and later adopters. The equal-family hazard coefficient is 0.936 ($p = 0.026$), with 851 country–family–years in 38 countries. Its market-cap counterpart is 0.912 ($p = 0.104$); the equal-family within-country coefficient is 0.104 ($p = 0.058$). This positive growth comparison is suggestive of learning from economic outcomes, but does not survive the full multiple-testing adjustment. It is different from evidence that disclosure itself caused growth or liquidity improvements.

4.2 Diffusion Mechanism 2 – Copying larger or related economies

We distinguish following a larger economy from responding to its outcomes. Geographic peers are fixed before the adoption period; recent and older implementations enter separately. Language-peer adoption provides a second familiarity measure. The estimates are not precise enough to es-

tablish a consistent copying effect across the two estimators and family weights (Online Appendix Tables OA2 and OA13). Legal source references to foreign standards establish the availability of reporting templates, but do not identify policymakers' motives.

4.3 Diffusion Mechanism 3 – Competition with economic rivals

The competition test uses implementation among countries with similar export patterns in 2000. Neither weight establishes a positive competitive response in the independent-choice models. Including coordinated European implementations changes the comparison; those estimates are retained separately. An externally coordinated legal wave is not an independent national response to rivals. Family-year effects and the independent-choice sample address different aspects of that distinction.

4.4 Diffusion Mechanism 4 – External pressure and financial dependence

The financing test relates implementation to prior aid and IMF credit relative to GDP. The within-country estimates do not establish a positive financing-pressure association. This aggregate proxy combines different financing relationships and does not identify a lender's conditions. Joint-channel models lack sufficient common support in the independent-choice sample. We retain those failed attempts alongside the fitted models rather than filling missing histories or reporting an unsupported joint mechanism.

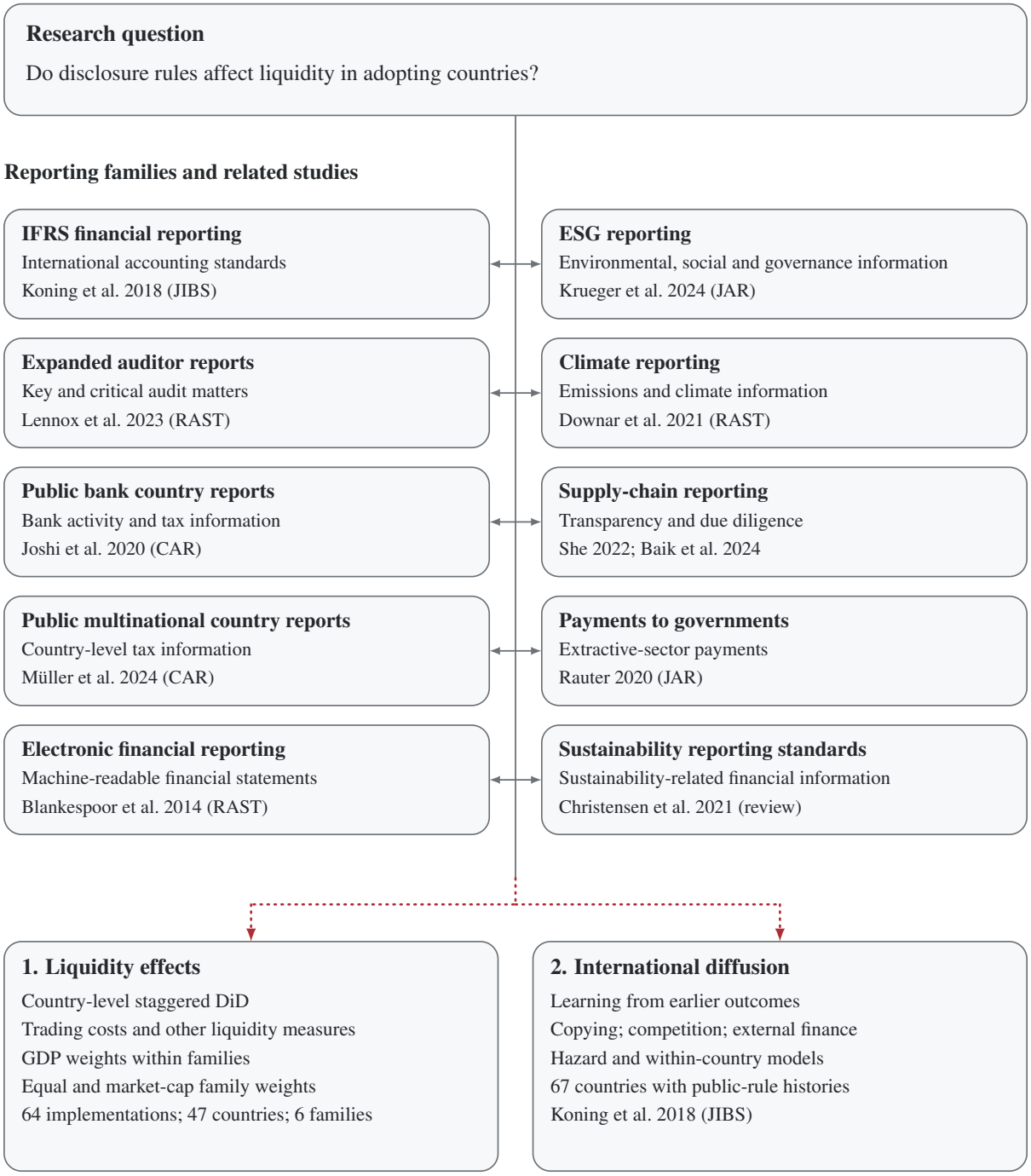
5 Conclusion

We bring public-disclosure requirements into a common country-level framework, extending the legal calendar with EU climate reporting and published country lists. The six-family liquidity comparison gives 10.2% lower trading costs under equal-family stacking and 17.1% under the market-cap family average, with different precision. The EU-inclusive climate comparison covers 14 adopting countries. These findings describe averages across heterogeneous reporting require-

ments and comparison samples, rather than a common benefit of every disclosure rule.

The implementation tests separate the presence of foreign rules from outcomes among earlier adopters. A positive growth-learning association appears in the equal-family hazard model, while the liquidity-learning interaction is imprecisely estimated. Neither establishes a diffusion mechanism after the full testing adjustment. The source-linked calendar makes these distinctions observable and provides a basis for more targeted issuer and policy comparisons.

Figure 1. Disclosure Rules, Liquidity and International Adoption



The diagram links the research question to ten reporting families and the two empirical comparisons. Citations identify related studies; Christensen et al. (2021) is a review, rather than a test of later ISSB mandates. The pooled liquidity sample supports six families and counts overlapping NFRD reforms once.

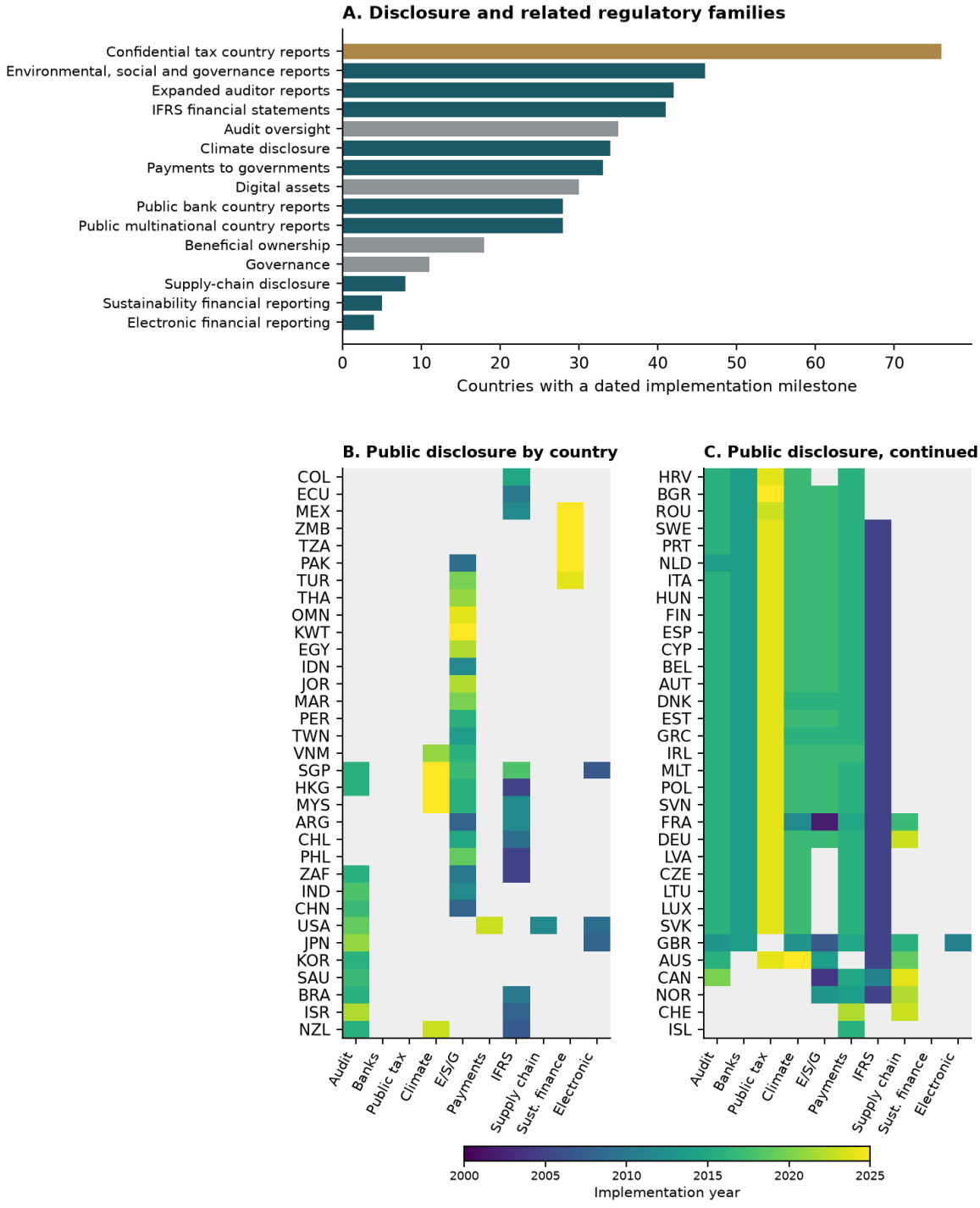


Figure 2. Policy families and international implementation

Panels show documented milestones dated 2000–2025 in the 90-jurisdiction universe. Country profiles are grouped by similarity in recorded public-disclosure families. Grey means no dated milestone in this calendar, not verified absence of reporting rules. This is not a complete global legal census; broader ESG rules can contain climate disclosures.

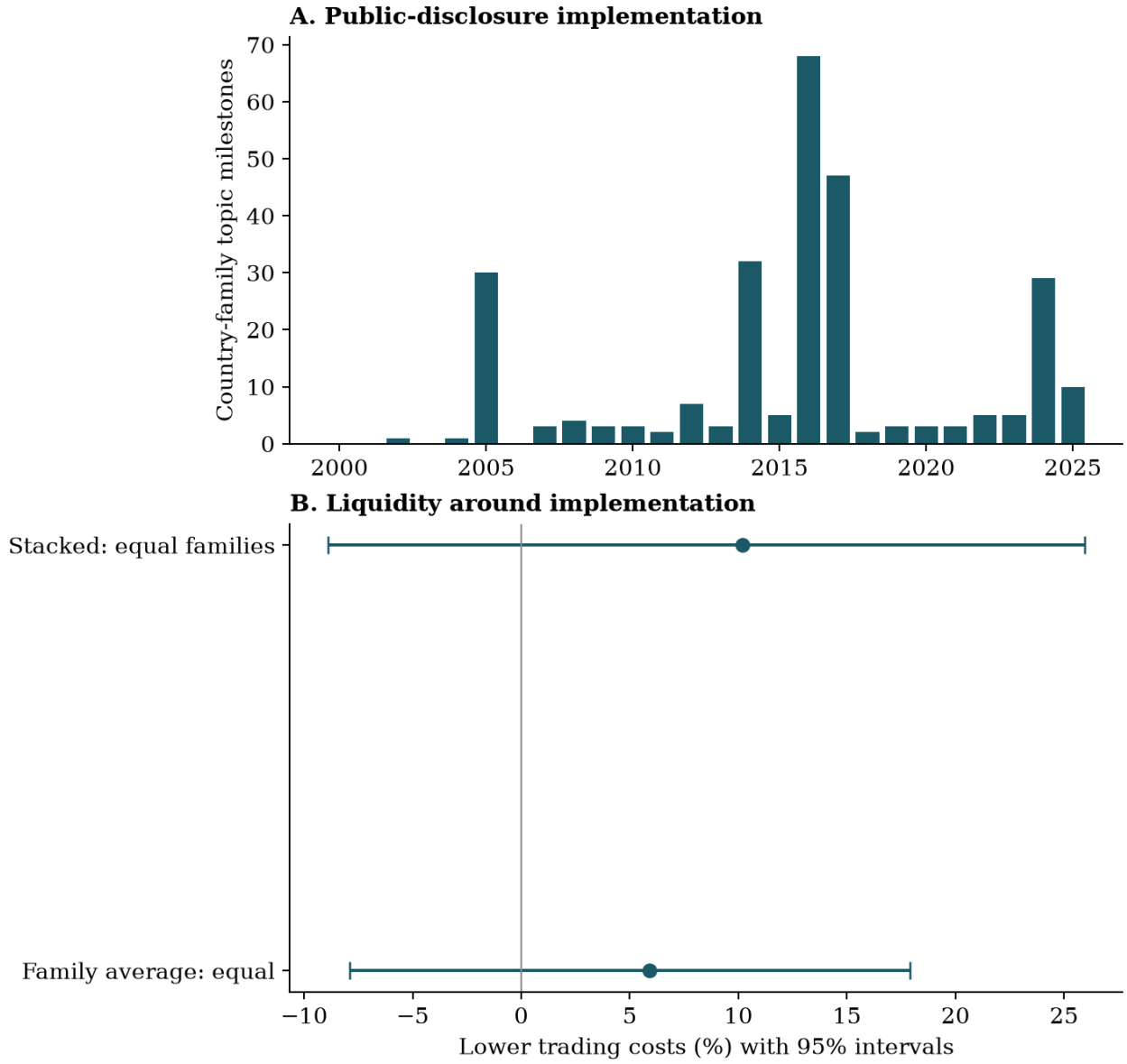


Figure 3. Public-disclosure implementation and stock liquidity

Panel A reports 269 country–family topic milestones, including overlapping topic tags. Panel B shows two equal-family estimates of lower trading costs, $100[1-\exp(\widehat{\beta})]$, with country-cluster 95% intervals. The supported liquidity sample contains 64 implementations and 47 countries. Positive values denote lower trading costs, not a proportionate increase in liquidity itself. Alternative family weights remain in Online Appendix Table OA6.

Table 1. Public-disclosure implementation milestones

Family	Countries	Liquidity adopters
Expanded auditor reports	42	20
Public bank country reports	28	13
Public multinational country reports	28	–
Climate disclosure	34	3
Environmental, social and governance reports	46	23
Payments to governments	33	–
IFRS financial statements	41	1
Supply-chain disclosure	8	4
Sustainability financial reporting	5	–
Electronic financial reporting	4	–

This table reports documented country–family implementation milestones dated 2000–2025, not counts of distinct laws. A country may appear in several families. Climate coverage includes EU NFRD materiality-based environmental reporting, not an unconditional mandate for every emissions metric. Duplicate NFRD climate tags are excluded from the pooled final column and reported separately in Table OA18. The final column counts adopting countries in the three-year liquidity design. A dash means the design lacks a supported cohort, not that the rule has no effect. Only the initial milestone selected for each country–family series is tabulated; amendments, successor phases, voluntary guidance and later schedules remain in the source archive. The ESG category includes mandated reports with different E, S and G content, materiality and disclosure-or-explanation provisions; it does not imply an unconditional mandate for every metric.

Table 2. Disclosure implementation and stock liquidity

	(1)	(2)
Family weights	Equal	Equal
Aggregation	Stacked	Average
Trading costs	−0.108	−0.061
	(0.096)	(0.068)
Trading-cost change	-10.2%	-5.9%
Country–stack–years	1704	1704
Countries	47	47
Families	6	6
Joint lead-test p	0.065	0.069

This table reports country-level rollout estimates. The outcome is log Bid-Ask Spread; lower values indicate lower trading costs and greater stock liquidity. Each cohort uses fixed companies in three pre-implementation years and years one to three after the documented milestone, excluding the implementation year. Country weights use GDP in the preceding year. The main table gives equal family mass. Market-cap alternatives appear in Online Appendix Table OA6. A stacked objective and an explicit average of family coefficients are distinct pooling targets. Standard errors in parentheses cluster by country and preserve dependence across policy stacks. *, ** and *** denote unadjusted significance at 10%,5% and 1%. The pool excludes a climate tag when the identical NFRD country-year reform is already represented under ESG; full climate-family estimates retain those cases. These are country-policy comparisons, not verified effects on every legally covered issuer.

Table 3. Foreign policy exposure and observed liquidity outcomes

	(1)	(2)
Estimator	Logit	Within
Family weights	Equal	Equal
Foreign-policy exposure	0.023	0.016
	(0.378)	(0.072)
Successful foreign adopters	−1.124	0.040
	(0.685)	(0.056)
Exposure x successful adopters	0.277	0.010
	(0.328)	(0.023)
Country–family–years	270	496
Countries	26	26
Implementations	28	29
Families	6	6

This table reports associations with documented implementation timing. Foreign-policy exposure is the lagged portfolio-weighted share of countries with the requirement. Successful foreign adopters are the share of observed earlier adopters with falling same-company log trading costs, measured before the focal implementation year; these are observed outcomes, not estimated effects caused by the foreign rule. Both primitives and their product enter together with lagged income, GDP growth and trade. Continuous primitives are standardized before the interaction. Logit includes family and year effects; Within denotes a linear probability model with country–family and family–year effects. Logit coefficients are log odds; within coefficients are probability changes. Country-cluster standard errors appear in parentheses. The samples differ because logit removes fixed-effect groups without outcome variation. Nominal stars use the conventions in Table 2; the full 160-model,256-focal-test adjustment is retained in the results files. Both family weights appear in Online Appendix Table OA12. Decision dates are not available for all cases, so these models do not establish learning at the political decision date.

A Variable Definitions

Table A1. Variables, Sources and Regression Terms

Variable or term	Definition and construction
Cleaned daily returns	Retain returns between -1 and 2. Exclude an adjacent-return pair when either daily return exceeds one and the compounded two-return change is below 0.2. The retained price is finite and at least US$0.10; a country trading date requires at least five securities with positive volume and sufficient price.
Country–family milestone, g	Source-supported implementation year for a country and family. The date-basis field distinguishes legal force, covered period, audit engagement and filing obligation. Unknown dates are not untreated observations.
Reporting family and audience	Ten public categories in Table 1. Confidential tax CbCR remains separate. A category does not establish universal issuer eligibility or mandatory disclosure of every item.
Treated country, D_{cg}	One for the implementing country in its family–cohort stack; zero for an eligible documented later adopter whose milestone follows the outcome window. Require at least one adopting and three comparison countries per stack.
Post-implementation, $Post_{gt}$	One in years $g+1$ to $g+3$ and zero in years $g-3$ to $g-1$. Omit year g. The shorter comparison uses two years on either side.
Treatment interaction	Product $D_{cg}Post_{gt}$. Event-time regressors replace Post with relative-year indicators. Year -1 is the reference; year zero is omitted.
Fixed-company sample	Select each company's largest positive-market-value security line in $g-1$. Retain the same company identifier, country and all four annual outcomes throughout the window. Require at least 30 companies per country–cohort.
Daily relative trading cost	Datastream quotes give $2(Ask-Bid)/(Ask+Bid)$ when $Bid \geq 0$, $Ask \geq Bid$ and $Ask+Bid>0$. The annual measure is the median of eligible daily observations.
Log trading costs	Log of the positive annual median relative trading cost, with at least 120 eligible observations. Winsorize positive annual medians at the year-specific first and ninety-ninth percentiles before logging. Country outcomes average company logs.
Dollar volume	US-dollar closing price times volume, with positive volume and a product of at least US$100. Missing price or volume remains missing.
Log price impact	Log of 10^8 times the annual median absolute cleaned daily return divided by dollar volume. Require at least 120 eligible observations and more than one distinct price; winsorize positive annual medians as above before logging.
Zero-return days	Fraction of observed cleaned daily returns equal to zero, with at least 120 eligible returns and more than one distinct price. Missing returns are not zeros. Country outcomes average company fractions.
Illiquidity index	Equal average of standardized log trading costs, log price impact and zero-return fraction. Calibration uses positive-cap, one-line-per-company observations in 2000. All components must exist. Higher values indicate less liquidity.
Country GDP weights	Constant-US-dollar GDP in $g-1$ from the archived World Bank series, normalized within stack. Cohort mass within family reflects the GDP represented by adopting countries.
Equal family mass	Give each supported family equal normalized objective mass. Residual treatment variation also affects a stacked coefficient; an explicit average uses family masses directly.

Variables, Sources and Regression Terms (continued)

Variable or term	Definition and construction
Market-cap family mass	Alternative mass from baseline sample market value represented by adopting-country companies. It is not a census of every covered issuer or all global market value. Liquidity and diffusion use their stated supported cases.
Fixed effects	Liquidity absorbs country and year effects within cohort stack. Diffusion logit includes family and year effects. Within-country diffusion absorbs country–family and family–year effects.
First implementation event	One in the documented country–family implementation year; zero in its documented pre-event risk years. Later cases can provide right-censored pre-event observations through 2025. Unknown histories remain missing.
Independent national choice	The primary diffusion sample excludes observations classified as coordinated obligations. The all-documented comparison retains them. This is a legal classification, not an estimated policymaker motive.
Portfolio exposure, P	Sum of prior-year portfolio weights on partners implementing by $t-1$. Use positive observed IMF total-economy asset positions, exclude self positions and normalize over partners with documented family histories.
Known-history portfolio coverage	Positive portfolio mass in documented-history partners divided by all observed positive foreign portfolio mass. The core tests require at least 50%. Unreported bilateral positions are not imputed zeros.
Foreign liquidity improvement	Negative change in mean company log trading costs from $t-2$ to $t-1$, holding companies and mapping fixed with at least 30 pairs. Positive means lower observed costs, not an estimated policy-caused gain.
Successful-adopter share, S	Unweighted share of observed eligible foreign adopters with positive liquidity improvement. Partners must implement by $t-2$; outcome-observed partners must cover at least 80% of eligible-adopter portfolio mass. No eligible adopters gives zero signal; missing required outcomes remain missing.
Successful-adopter exposure, *Lexp*	Sum of portfolio weights on eligible earlier adopters with positive observed liquidity improvement. This weighted exposure differs from the unweighted share S.
Learning interaction, $P \times S$	Product of standardized exposure and success primitives, with both lower-order terms included. The logit coefficient is a log-odds interaction, not a percentage-point probability effect.
Larger peer, B	Whether the first larger-GDP peer among fixed geographic neighbours implemented by $t-1$. Fix size with GDP in 2000. An absent peer or undocumented history is not coded zero.
Recent and older copying	For that same larger peer, $Bnew = 1$ if implementation is in $t-1$; $Bold = 1$ if earlier than $t-1$. Retain the documented-history requirement.
Language-peer implementation	Prior-year adoption share among official-language peers with documented histories; require at least three observed peers. The proxy does not establish actual copying.
Trade-competitor implementation	Prior adoption share among export-pattern peers selected with the archived 2000 network. Require at least three documented histories; do not reselect peers based on current outcomes.
Growth learning	Indicator that median prior-year GDP growth among earlier adopters exceeds the median among later adopters in the ten nearest documented peers. Require at least 80% macro coverage in each group. No earlier adopters gives zero; if all peers adopted, compare their median with focal-country growth. Other missing support remains missing.
External financial dependence	Archived prior-year aid plus IMF credit and SDR allocations relative to GDP. The proxy does not identify actual financing conditions imposed by lenders.

Variables, Sources and Regression Terms (continued)

Variable or term	Definition and construction
Income per capita	Lagged log positive constant-price GDP per capita from the archived World Bank series; included in diffusion and controlled liquidity models.
GDP growth	Lagged annual GDP-growth percentage from the archived country series. Missing values are not replaced by assumed growth.
Trade openness	Lagged exports plus imports as a percentage of GDP from the archived World Bank series.
Standardized primitives	Within each diffusion sample, centre and scale nonbinary primitives and controls using population standard deviations. Binary variables retain their scale. Build interactions afterwards. Paired weights use identical samples and scaling.
Joint channel model	Exposure, successful-adopter exposure, larger-peer adoption, trade peers, language peers and financial dependence, with three country controls; estimated only on a supported data intersection.
Support and inference	Diffusion requires at least 50 rows, 15 countries, 10 events and 10 nonevents, suitable rank and variance. Logit drops fixed-effect groups without event variation. Standard errors cluster by country; nominal inference uses $G-1$ degrees of freedom.
Lead test and adjusted p-values	Liquidity uses a joint country-cluster Wald test of pre-event coefficients. Diffusion Holm adjustment covers 256 focal attempts in 160 fits; unestimated attempts receive $p=1$ for adjustment only.
Restricted five-model benchmark	Historical 30-implementation estimates retain stacked equal/cap mass, explicit family averages and covariance-weighted GLS. They are distinct historical benchmarks, not estimates on the expanded sample.
Pooled-eligible implementation	A documented country–family milestone retained for pooling. A climate tag is excluded when the same NFRD country-year reform is represented under ESG. Full climate-family estimates retain those topic tags; histories remain available for peer signals.

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